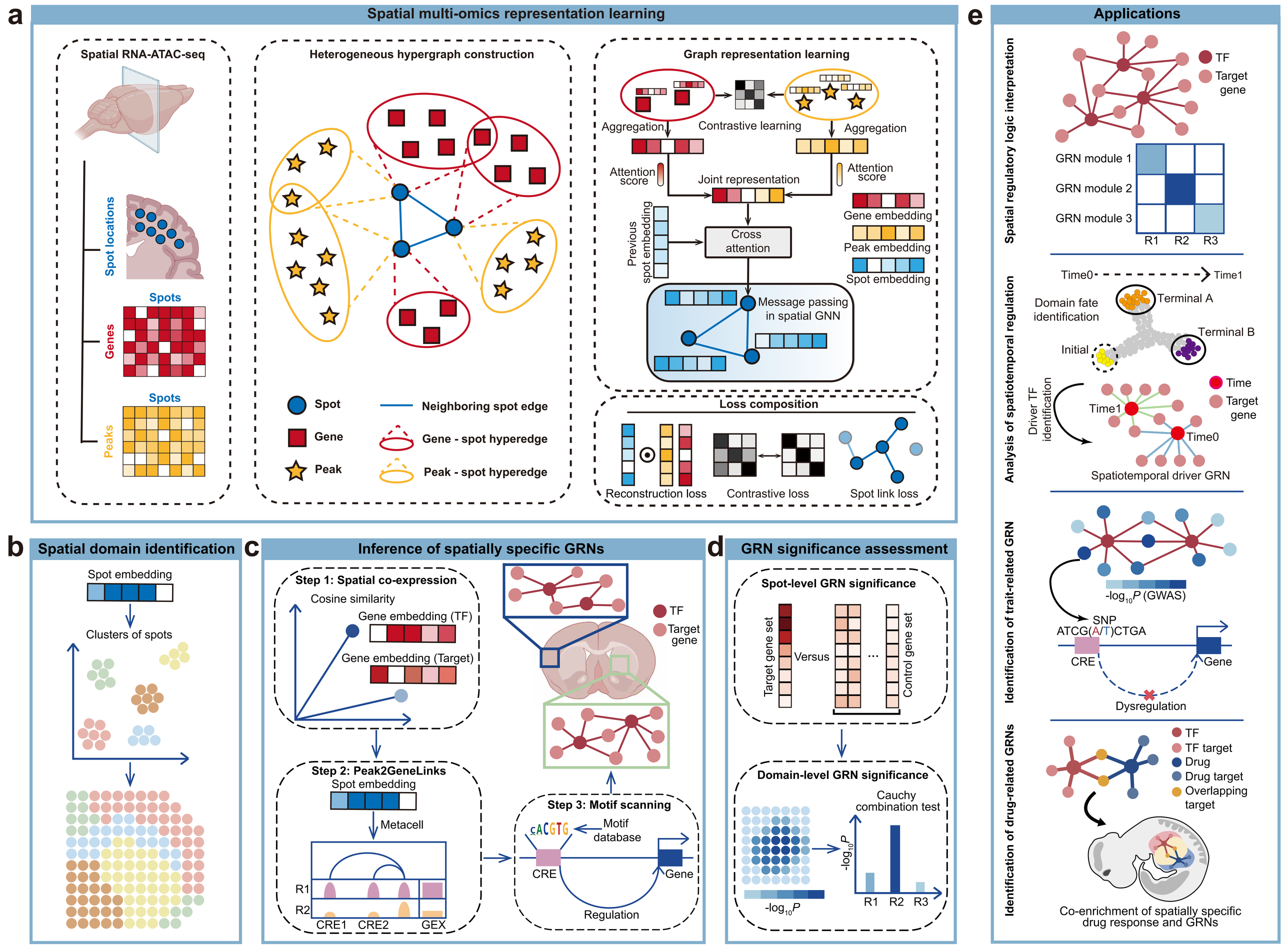
a
Spatial multi-omics representation learning
Spatial RNA-ATAC-seq
Spot locations
Spots
Genes
Spots
Peaks
Heterogeneous hypergraph construction
Spot
Gene
Peak
Neighboring spot edge
Gene - spot hyperedge
Peak - spot hyperedge
Graph representation learning
Aggregation
Contrastive learning
Aggregation
Attention score
Joint representation
Attention score
Previous spot embedding
Cross attention
Gene embedding
Peak embedding
Spot embedding
Message passing in spatial GNN
Loss composition
Reconstruction loss
Contrastive loss
Spot link loss
b
Spatial domain identification
Spot embedding
Clusters of spots
c
Inference of spatially specific GRNs
Step 1: Spatial co-expression
Cosine similarity
Gene embedding (TF)
Gene embedding (Target)
TF
Target gene
Step 2: Peak2GeneLinks
Spot embedding
Metacell
R1
R2
CRE1
CRE2
GEX
Step 3: Motif scanning
cACGTG
Motif database
CRE
Gene
Regulation
d
GRN significance assessment
Spot-level GRN significance
Target gene set
Versus
Control gene set
Domain-level GRN significance
Cauchy combination test
-log10P
R1
R2
R3
e
Applications
Spatial regulatory logic interpretation
TF
Target gene
GRN module 1
GRN module 2
GRN module 3
R1
R2
R3
Analysis of spatiotemporal regulation
Time0
Time1
Domain fate identification
Terminal A
Terminal B
Initial
Driver TF identification
Time
Target gene
Time1
Time0
Spatiotemporal driver GRN
Identification of trait-related GRN
-log10P (GWAS)
SNP
ATCG(A/T)CTGA
CRE
Gene
Dysregulation
Identification of drug-related GRNs
TF
TF target
Drug
Drug target
Overlapping target
Co-enrichment of spatially specific drug response and GRNs